

E-Commerce Sales Data Analytics

Project Overview

End-to-end analysis of an e-commerce transactions to uncover revenue drivers, product performance, and marketing ROI. Built in Excel with Pivot Tables, Summary Stats, and Visualizations.

Key Objectives

1. Identify top-selling products and months
2. Analyze which referral channels drive the most revenue
3. Quantify the relationship between Sales and Profit
4. Clean and structure raw order data for analysis

Tools Used

- Excel: Data cleaning, Pivot Tables, SUMIFS, Correlation, Charts.
- Skills: Data cleaning, EDA, Business KPIs, Visualization.

Key Findings

| Metric | Result |

| Total Sales | \$489,087.50 |

| Average Order Value | \$1,056.34 |

| Total Orders | 463 |

| Top Product | Printer: \$88,736.92 |

| Best Referral Source | Instagram: \$112,552.06 |

| Peak Month | June: \$66,383.46 |

| Sales-Profit Correlation | 0.99 - Very strong positive link |

Product Performance

1. Printer - \$88,736.92
2. Desk - \$75,912.24
3. Tablet - \$75,264.90
4. Laptop - \$68,235.96

5. Chair - \$68,185.29

Referral Channel Performance

1. Instagram - \$112,552.06

2. Google - \$103,688.01

3. Email - \$100,291.35

4. Referral - \$96,285.00

5. Facebook - \$76,271.08

Monthly Trend

Peak: June \$66,383.46 | Lowest: July \$26,592.97

Data Cleaning Steps

1. Removed duplicates + nulls from Cleaned Data sheet

2. Standardized Order Status : Delivered, Cancelled, Returned, Pending, Shipped

3. Created derived columns: Total Price, Profit, Month for analysis

4. Handled 1 outlier in the dataset

Insights & Recommendations

1. Double down on Printers: Highest revenue product category.

2. Scale Instagram Marketing: It's the first referral source by revenue.

3. Plan for June Spikes: Seasonal increase suggests Q2 campaigns work.

4. Profit Focus: 0.99 correlation means increasing sales directly boosts profit.

5. Next Step: Rebuild this in Power BI/SQL for interactive dashboards and DAX.

Project Files

Esther Udoh Decodelabs Project 2 .xlsx

README.md

Author

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Passionate about solving real-world business problems